

53- Bayesian Zone Collapse When a Fix Arrives.

Purpose: Our autonomous ocean platform may not know the exact location of a drifting float, underwater glider, or observation target.

Initially, we have a large predicted location area. When a GPS or acoustic position fix arrives, the uncertainty becomes much smaller.

This is called Bayesian updating.

Before Basic concept:

- Before receiving the fix: We have a forecast of where the object may be.
- After receiving the fix: We combine the forecast and the new measurement.
- Result: The probable location zone becomes smaller.

This helps the autonomous platform decide:

- Where should the glider Search?
- Where should the surface vehicle travel?
- Where should the communication gateway wait?
- How can we reduce unnecessary movement and energy consumption?

1. Formula:

~~Think about it~~

$$P(x/z) \propto P(z/x) p(x)$$

Posterior = New information combined with ~~prior~~ prior information.

For Gaussian uncertainty, the standard deviation is updated using:

$$\sigma_{\text{posterior}} = \left(\frac{1}{\sigma_{\text{forecast}}^2} + \frac{1}{\sigma_{\text{fix}}^2} \right)^{-1/2}$$

Where, σ_{forecast} = uncertainty before receiving the new fix

σ_{fix} = Uncertainty of the GPS/acoustic measurement.

$\sigma_{\text{posterior}}$ = Final uncertainty after combining both

2. Calculation:

- Forecast uncertainty $\approx 24 \text{ km}$
- GPS burst uncertainty $\approx 0.1 - 0.5 \text{ km}$

Case - A: GPS fix uncertainty = 0.5 km

given, $\sigma_{\text{fix}} = 24 \text{ km}$

$\sigma_{\text{fix}} = 0.5 \text{ km}$

$$\sigma_p = \left(\frac{1}{\sigma_f^2} + \frac{1}{\sigma_{\text{fix}}^2} \right)^{-1/2}$$

$$= \left(\frac{1}{(24)^2} + \frac{1}{(0.5)^2} \right)^{-1/2} = \left(\frac{1}{576} + \frac{1}{0.25} \right)^{-1/2}$$

$$= (0.001736 + 4)^{-1/2} = (4.001736)^{-1/2}$$

$$\sigma_p \approx 0.4999 \text{ km}$$

$\therefore$ The uncertainty reduces from approximately 24 km $\rightarrow$ 0.5 km

Case-B: GPS fix uncertainty = 0.1 km

$$\sigma_p = \left(\frac{1}{24^2} + \frac{1}{0.1^2} \right)^{-1/2}$$

$$= \left(\frac{1}{576} + \cancel{100} \right)^{-1/2} = (100.001736)^{-1/2}$$

$$\sigma_p \approx 0.1 \text{ km}$$

$$0.1 \text{ km} = 100 \text{ meters}$$

$\therefore$ The updated uncertainty is approximately 100 meters.

56 - Pump Energy Budget - Buoyancy Engine

Purpose: Our autonomous underwater platform needs to change its depth.

Instead of continuously using a propeller, it can use a buoyancy engine.

The buoyancy engine changes the vehicle's volume or displaces water using a pump.

- Increase ~~buoyant~~ displacement $\rightarrow$ more buoyancy $\rightarrow$ Vehicle rises.

- Decreases displacement $\rightarrow$ less buoyancy $\rightarrow$ Vehicle sinks.

This is useful because the vehicle can move vertically with relatively low energy compared with continuous powered propulsion.

1. Basic physics:

The pump must move a certain volume of fluid against pressure.

The mechanical work is:

$$W_{\text{mech}} = P \Delta V$$

where, W_{mech} = Mechanical work required
 P = Pressure at the operating depth
 ΔV = Volume of fluid moved

2. Pressure at 2000 metres

Given, $P \approx 200 \text{ bar}$

converting bar to pascals:

$$1 \text{ bar} = 100,000 \text{ Pa}$$

$$\therefore 200 \text{ bar} = 200 \times 100,000$$

$$\Rightarrow \boxed{P = 20,000,000 \text{ Pa} = 20 \text{ MPa}}$$

3. Convert Volume:

Given, $\Delta V = 0.7 \text{ L}$

Since, $1 \text{ L} = 0.001 \text{ m}^3$

$$0.7 \text{ L} = 0.0007 \text{ m}^3$$

4. Calculate mechanical work

$$W_{\text{mech}} = P \Delta V$$

$$W_{\text{mech}} = 20,000,000 \times 0.0007$$

$$\boxed{W_{\text{mech}} = 14,000 \text{ J}}$$

(or)

$$\boxed{W_{\text{mech}} = 14 \text{ kJ}}$$

Converting Joules to watt-hours

$$1 \text{ Wh} = 3,600 \text{ J}$$

$$E_{\text{mech}} = \frac{14,000}{3,600}$$

$$\boxed{E_{\text{mech}} \approx 3.89 \text{ Wh}}$$

Ideally, the pump requires approximately 3.9 Wh of mechanical energy.

5. Account for pump efficiency

A real pump is ~~at~~ not 100% efficient.

Some energy is lost due to:

- Friction
- Heat
- Motor losses
- Hydraulic losses
- Leakage
- Pressure regulation.

Assume pump efficiency:

$$\eta_{\text{pump}} = 0.3 \text{ to } 0.4$$

The electrical energy required is:

$$E_{\text{elec}} = \frac{E_{\text{mech}}}{\eta_{\text{pump}}}$$

At 40% efficiency,

$$E_{\text{elec}} = \frac{3.89}{0.4}$$

$$E_{\text{elec}} \approx 9.72 \text{ Wh}$$

At 30% efficiency,

$$E_{\text{elec}} = \frac{3.89}{0.3}$$

$$E_{\text{elec}} \approx 12.96 \text{ Wh}$$

$\therefore$ the ideal calculated electrical requirement is approximately: $9.7 - 13.0 \text{ Wh}$

6. Recharge - window calculation

Delivering 28 Wh takes approximately 67 mins at 25W, 42 minutes at 40W, and 22 minutes at 75W.

Formula:

$$t = \frac{E}{P}$$

Where, $t \rightarrow$ time in hours

$E \rightarrow$ energy in Wh

$P \rightarrow$ Power in watts

At 25W,

$$t = \frac{28}{25} = 1.12 \text{ hours}$$

Convert to minutes:

$$1.12 \times 60 = 67.2 \text{ minutes}$$

$$t \approx 67 \text{ minutes}$$

At 40W,

$$t = \frac{28}{40} = 0.7 \text{ hours}$$

$$0.7 \times 60 = 42 \text{ minutes}$$

$$t \approx 42 \text{ minutes}$$

At 75W,

$$t = \frac{28}{75} = 0.3733 \text{ hours}$$

$$0.3733 \times 60 \approx 22.4 \text{ minutes}$$

$$t \approx 22 \text{ minutes}$$

39 - Data and Energy Budget

Purpose:

Our autonomous platform collects ocean data

Such as:

- Temperature
- Salinity
- Pressure
- Conductivity
- Depth
- Possibly dissolved oxygen or other sensor measurements

The challenge is, "How can we send this data to researchers while keeping the system low-cost and energy-efficient?"

Directly transmitting every profile through a satellite modem can consume more energy and cost more than using a nearby local communication link.

1. Formula:

$$E_{\text{comm}} = (\text{data} \times \text{energy per KB}) + P_{\text{handset}} t_{\text{session}}$$

where,

data $\rightarrow$ amount of information transmitted
Energy per KB $\rightarrow$ Energy needed to send each kilobyte.

P_{handset} $\rightarrow$ Power consumed by the communication session.

$E_{comm} \rightarrow$ Total communication energy

Assume, One profile contains;

$$D = 100 \text{ kB}$$

Communication energy is;

$$e_{kB} = 0.002 \text{ Wh/kB}$$

Then,

$$E_{data} = D \times e_{kB} \\ = 100 \times 0.002$$

$$E_{data} = 0.2 \text{ Wh}$$

If the modem consumes;

$$P_{handset} = 5 \text{ W}$$

and communication session lasts;

$$t_{\text{session}} = 2 \text{ minutes}$$

$$\text{time} \rightarrow \text{hours}; \quad t = \frac{2}{60} = 0.0333 \text{ hours}$$

Session energy:

$$E_{comm} = E_{data} + E_{\text{session}}$$

$$= 0.2 + 0.167$$

$$E_{comm} = 0.367 \text{ Wh per profile}$$

Satellite energy comparison:

Assume,

$$E_{\text{direct}} = 5 \text{ Wh}$$

Let's calculate the potential saving if the gateway reduces satellite energy by 80%.

Gateway routing

An 80% reduction means:

$$E_{\text{gateway}} = 20\% \times E_{\text{direct}}$$

$$= 0.2 \times 5$$

$$E_{\text{gateway}} = 1 \text{ Wh}$$

Energy saved,

$$E_{\text{saved}} = E_{\text{direct}} - E_{\text{gateway}}$$
$$= 5 - 1$$

$$E_{\text{saved}} = 4 \text{ Wh}$$

Full example: One observation profile

Profile size = 100KB, Local data transfer energy = 0.2 Wh

Gateway satellite forwarding energy = 0.8 Wh

Total gateway energy = local + satellite forwarding

Then,

$$E_{\text{gateway}} = E_{\text{local}} + E_{\text{satellite, forward}}$$

$$E_{\text{gateway, total}} = 0.2 + 0.8 = 1.0 \text{ Wh}$$

Assume direct satellite transmission would require:

$$E_{\text{direct}} = 5 \text{ Wh}$$

Then,

$$E_{\text{saved}} = 5 - 1 = 4 \text{ Wh}$$

Percentage saving:

$$\text{Saving} / \% = \frac{E_{\text{direct}} - E_{\text{gateway}}}{E_{\text{direct}}} \times 100 = \frac{5 - 1}{5} \times 100$$

$$\text{Saving} \% = 80\%$$